

Modeling Decisions — Roomies Relational Domain Model

1. Entities introduced that were not named explicitly

- **shared_spaces + property_shared_spaces** — shared spaces are an attribute in the brief, not an entity. Modeled as catalog instead of free text. Unlike neighborhoods and amenities, no user story assigns a catalog, we extend moderator management to it by analogy following the same logic.
- **property_amenities** — M:N connection between properties and amenities. Written as an explicit join table
- **listing_photos** — there can be multiple photos per listing, each with different data
- **saved_listings** — Created for the purpose of saving across multiple sessions, works as a M:N connection between listings and seekers by saving

2. Representing the listing and application lifecycles

Both lifecycles are modeled as a single **status** enum column on the entity itself, rather than a separate state-history table

- **Listing:**
 1. draft: Host creating a listing
 2. published: Listing on page visible by seekers and visitors
 3. reserved: An application has been accepted
 4. rented: The applicant has moved in the property
 5. withdrawn: Host or moderator takes down the listing from the page
- **Application:**
 1. pending: Host has not seen or interacted with the application
 2. shortlisted: Host interested with application and wants to see seeker (could trigger visit)
 - a. accepted: Host has accepted the applicant for stay (Cancels all application remaining)
 - b. rejected: The host declined the application
 - c. withdrawn: The seeker canceled the application themselves

3. Domain assumptions not settled by the project description

- Host/seeker are not roles but positions derived from ownership (host = owns the property behind a listing) and visitors are the lack of a role, not another one. user_role only distinguishes member vs. moderator.
- Amenities are property-level (e.g. elevator belongs to the building); is_furnished and has_private_bathroom are room-level columns on listings, not amenities.
- Reviews attach to the property (per the brief) but are anchored to a visit, so 'one review per completed visit' becomes a plain unique constraint on visit_id.
- An application can have several visits (reschedule/cancel creates a new visit row rather than mutating the old one).
- Neighborhood names are unique per city, not globally.
- deposit defaults to 0 and minimum_stay_days is nullable — both are optional terms a host may choose not to impose.
- If applications are cancelled or withdrawn instead of being “voided”
- The rating in a review goes from 1 to 5
- Shared spaces are not amenities, they are their own catalog manageable by moderators
- Reports have their own reviewer with its review date, they also have to have a reason
- Visits are not mandatory on applications, but every visit connects to one.